

Customer Repeat Purchase Analysis

Measuring loyalty and identifying repeat-customer segments in the Online Retail II dataset

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1,500

Unique customers

52.1%

Repeat rate

3,931

Total orders

45 days

Avg. time to 2nd order

1. Executive Summary

This report measures repeat-purchase behaviour in the Online Retail II transaction data and segments customers by loyalty. Of **1,500** identified customers, **781 (52.1%)** placed two or more orders, while **47.9%** bought only once. Loyal and Regular customers (4+ orders) make up only **22.8%** of the base but generate **55.4%** of revenue. Average order value is almost identical for first-time (**77.98**) and repeat (**77.59**) orders, so loyalty is driven by purchase **frequency**, not basket size.

2. Objective and Scope

Objective: Understand loyalty metrics. **Deliverables:** repeat rate, segment table and insights. A **repeat customer** is defined as a customer with **2 or more unique invoices** in the analysis period.

3. Data and Method

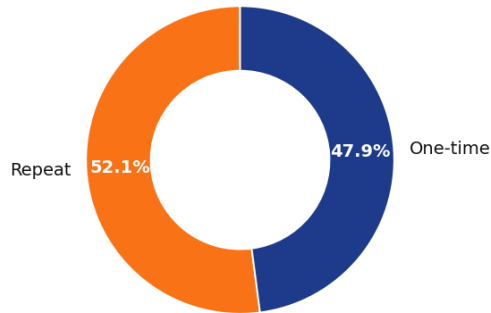
Dataset	Online Retail II (Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country)
Period	07 Dec 2009 to 09 Dec 2011
Rows	18,810 raw rows, 17,745 after cleaning
Cleaning	Removed 710 rows with missing Customer ID, 355 cancellation rows (invoice starting with 'C'), and rows with non-positive quantity or price
Method	Python (pandas): invoices grouped per customer, orders numbered by date, customers segmented by order count, AOV compared for first vs later orders

4. Findings

4.1 Repeat rate

Repeat rate = repeat customers / total customers = 781 / 1,500 = 52.07%. Roughly half of customers never return after their first order, which is the largest retention gap in the data.

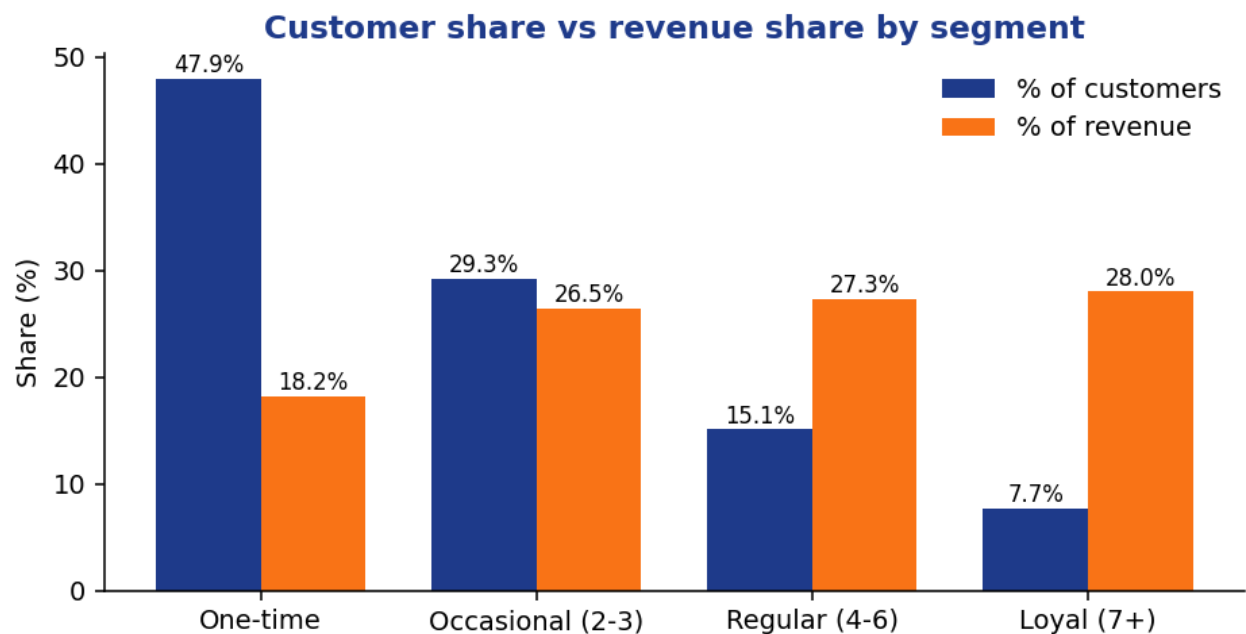
Repeat vs one-time customers



4.2 Customer segment table

Segment	Customers	% Cust.	Avg orders	Avg AOV	Revenue	% Rev.
One-time	719	47.9%	1.00	77.27	55,556	18.2%
Occasional (2-3)	439	29.3%	2.39	77.06	80,879	26.5%
Regular (4-6)	227	15.1%	4.75	77.62	83,458	27.3%
Loyal (7+)	115	7.7%	9.44	78.63	85,699	28.0%
Total	1,500	100%	2.62	77.74	305,592	100%

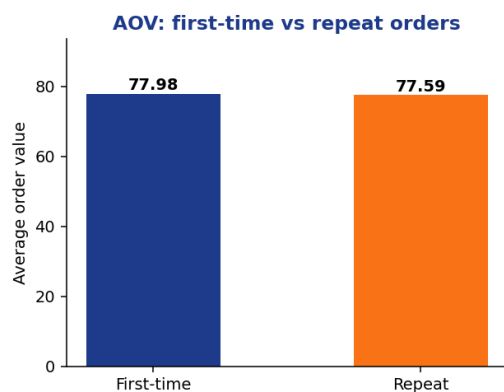
Avg AOV is the mean of each customer's average order value. Revenue is in the dataset's currency (GBP).



Customers with 4+ orders are **22.8%** of the base but contribute **55.4%** of revenue, while one-time buyers are **47.9%** of customers and only **18.2%** of revenue.

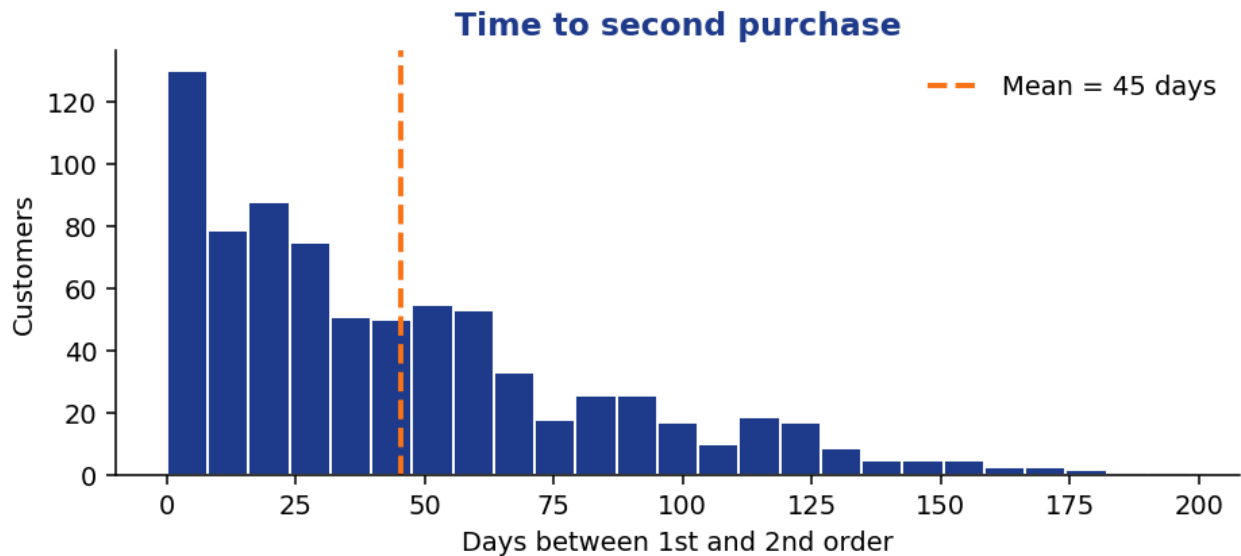
4.3 First-time vs repeat order value

Order type	Orders	Mean AOV	Median AOV
First-time order	1,500	77.98	64.22
Repeat order	2,431	77.59	64.35



Repeat orders average **77.59** versus **77.98** for first orders, a difference of **-0.5%**. Customers do not spend more per order as they become loyal. Their value grows because they order more often.

4.4 Time to second purchase



Among the 781 customers who ordered at least twice, the second order came on average **45 days** (median **34**) after the first. This sets the window for follow-up campaigns.

5. Key Insights

- **Retention gap:** only 52.1% of customers repeat; 47.9% buy once and leave.
- **Revenue concentration:** Regular and Loyal customers are 22.8% of the base but give 55.4% of revenue.
- **Loyal core:** customers with 7+ orders are 7.7% of customers and 28.0% of revenue.
- **Frequency over basket size:** repeat AOV (77.59) is nearly equal to first-order AOV (77.98).
- **Purchase cycle:** the second order arrives about 45 days after the first.

6. Recommendations

- Trigger a personalised follow-up offer around day 29-40 after the first purchase, before the typical second-order window closes.
- Launch a loyalty or VIP programme for Regular and Loyal customers to protect the revenue they generate.
- Test bundles, free-shipping thresholds and cross-selling to lift basket size, since AOV does not rise with loyalty.
- Track repeat rate and time-to-second-order monthly as core retention KPIs.

Conclusion

Only **52.1%** of customers made a repeat purchase, while **47.9%** bought just once, leaving a large loyalty gap. Loyal customers (7+ orders) are a small share of the base but generate **28.0%** of total revenue, and their order value is similar to first-time buyers, so loyalty is driven by purchase frequency, not basket size. The business should convert one-time buyers with a targeted follow-up offer within about **45 days** of the first order and reward its Regular and Loyal segments to protect core revenue.